21ASC305T
ROCKET PROPULSION

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- ***Unit-1 - Basic Concepts 9 Hour***
- *Rocket propulsion systems -working principle, Classification of rockets, Rocket equation, Rocket Nozzles and its classifications -Nozzle performance- equilibrium and frozen flow, Characteristic velocity and Thrust coefficient, Performance parameters and efficiencies of rocket, Staging and clustering of rocket*
- ***Unit-2 - Chemical Propellants 9 Hour***
- *Chemical propellants -Molecular mass, Specific heat capacities, Specific heat ratio, mixture ratio, Stoichiometric ratio and equivalence ratio, Energy release during combustion-Heat of formation and Heat of combustion-Combustion Instabilities-Criterion for choice of propellant, Solid Propellants-Composition and processing, Liquid Propellants-Classifications of liquid propellant, Storability of liquid propellant, Chemical Equilibrium*
- ***Unit-3 - Solid Rocket Engine 9 Hour***
- *Propulsion Elements for Solid Rocket Motors, Solid propellant combustion and Internal Ballistics of Motors - Mechanism of burning, Ignition System-Igniter types - Solid propellant grain design - Burn Rate-Factors influencing burn rates, burn rate index for stable operation, Action time and burn time, Design of Solid Propellant rocket, Simulation of solid rocket motor performance.*
- ***Unit-4 - Liquid Rocket Engine 9 Hour***
- *Liquid Propellant Rocket-Hardware components and its functions, Propellant feed systems –Gas pressure feed system, Turbo pump feed systems and engine cycles, Tank Pressurization - Types of injectors – Thrust chamber, Cooling of Thrust chamber - Cryogenic propulsion system and its special features, Simulation of Liquid rocket performance, Hybrid rockets- Review of solid-fuel regression rate behavior in classical and non-classical Hybrid Rocket Motors - Rocket engine design using software.*
- ***Unit-5 - Non-conventional Propulsion Techniques 9 Hour***
- *Non - conventional propulsion techniques -Electrical Rockets-Electro-thermal propulsion system-Arc-jet thruster, Resistojet Thruster-Electro-static thruster-Ion thruster, Bombardment Ionization, Hall Effect Thruster -Electro-magnetic propulsion system-Magneto plasma dynamic accelerator, Pulsed Plasma accelerator- Laser Propulsion -Nuclear powered rockets- Solar Propulsion system*

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- *1. George P. Sutton, Oscar Biblarz, "Rocket propulsion elements", Wiley India Pvt Ltd. Ninth Edition, 2017*
- *2. Ramamurthi.K, "Rocket propulsion", Laxmi Publications, India, Second edition, 2019.*
- *3. J D. Mattingly and K M. Boyer,"Elements of Propulsion: Gas Turbines and Rockets", Second Edition,2016*
- *4. Philip Hill and Carl Peterson, "Mechanics and thermodynamics of propulsion", Pearson India, second edition, 2010.*
- *5. Stephen R. Turns, "An Introduction to Combustion: Concepts", McGraw-Hill Education; 4th edition, 2020*
- *6. Yahya S M, "Gas Tables",8th Edition, 2018*

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Unit-3 - Solid Rocket Engine

- *Propulsion Elements for Solid Rocket Motors, Solid propellant combustion and Internal Ballistics of Motors -Mechanism of burning, Ignition System-Igniter types - Solid propellant grain design - Burn Rate-Factors influencing burn rates, burn rate index for stable operation, Action time and burn time, Design of Solid Propellant rocket, Simulation of solid rocket motor performance.*

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PROBLEM

- An end-burning rocket uses a cylindrical double-base propellant grain with a diameter of 200 mm and generates a thrust of 350 N over a period of 300 s. The thrust coefficient is 1.15. The characteristics of the propellant are:
- Density of propellant grain = 1500 kg/m^3
- a = 4 mm/s; n = 0.5
- C* =1500 m/s
- Determine: (i) the length of the grain
- (ii) throat diameter of the nozzle

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GIVEN DATA

- Grain diameter, $D = 200$ mm $= 0.2$ m
- Thrust, $F = 350$ N
- Burning time, $t_b = 300$ s
- Thrust coefficient, $C_F = 1.15$
- Propellant density, $\rho_p = 1500$ kg/m^3
- Burning-rate coefficient, $a = 4$ mm/s $= 0.004$ m/s
- Pressure exponent, $n = 0.5$
- Characteristic velocity, C∗=1500 m/s

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Propellant mass flow rate

$$\dot{m} = \frac{F}{C_F C^*}$$

$$\dot{m} = \frac{350}{1.15(1500)}$$

$$\boxed{\dot{m} = 0.2029 \text{ kg/s}}$$

Total propellant consumed

$$m_p = \dot{m} t_b$$

$$m_p = 0.2029(300) = 60.87 \text{ kg}$$

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Length of the grain:

For an end-burning cylindrical grain, the **burning area** is the circular end area:

$$A_b = \frac{\pi D^2}{4}$$

$$A_b = \frac{\pi (0.2)^2}{4} = 0.031416 \text{ m}^2$$

The grain volume is

$$V_p = A_b L$$

$$m_p = \rho_p A_b L$$

$$L = \frac{m_p}{\rho_p A_b}$$

$$L = \frac{60.87}{1500(0.031416)}$$

$$\boxed{L \approx 1.29 \text{ m}}$$

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Using burning rate, the length of grain is

$$\dot{m} = \rho_p A_b r_b$$

$$r_b = \frac{\dot{m}}{\rho_p A_b}$$

$$r_b = \frac{0.2029}{1500(0.031416)} = 0.004307 \text{ m/s}$$

$$\boxed{r_b = 4.307 \text{ mm/s}}$$

$$t_b = \frac{L}{r_b}$$

$$L = r_b t_b = (0.004307)(300)$$

$$\boxed{L = 1.292 \text{ m}}$$

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Throat diameter

Chamber pressure using the burning-rate law

$$r_b = a p_c^n$$

$$4.307 = 4 p_c^{0.5}$$

$$\sqrt{p_c} = \frac{4.307}{4} = 1.0768$$

$$p_c = (1.0768)^2$$

$$\boxed{p_c = 1.159 \text{ MPa}}$$

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$$F = C_F p_c A_t$$

$$A_t = \frac{F}{C_F p_c}$$

$$A_t = \frac{350}{1.15(1.159 \times 10^6)}$$

$$A_t = 2.628 \times 10^{-4}\ \text{m}^2$$

$$A_t = \frac{\pi d_t^2}{4}$$

$$d_t = \sqrt{\frac{4A_t}{\pi}}$$

$$d_t = \sqrt{\frac{4(2.628 \times 10^{-4})}{\pi}}$$

$$\boxed{d_t \approx 0.0183\ \text{m}}$$

$$\boxed{d_t \approx 18.3\ \text{mm}}$$

Problem

- **An end-burning solid propellant rocket, having a throat diameter of 28 mm, is employed as the sustainer of a missile. It needs to provide a constant thrust of 5 KN. If a double-base propellant with a burn rate law $r = 0.004P^{0.45}$ where P is in Pa and r is in mm/s is used, determine the diameter of the propellant grain.** You can assume the specific impulse of the booster to be 1900 N.s/kg. The C^* and density of the double-base propellant are 1600 m/s and 1540 kg/m^3 respectively.

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Chamber pressure:

Mass flow rate

$$\dot{m} = \frac{F}{I_{sp}}$$

$$\dot{m} = \frac{5000}{1900}$$

$$\boxed{\dot{m} = 2.632 \text{ kg/s}}$$

$$C^* = \frac{p_c A_t}{\dot{m}}$$

$$p_c = \frac{\dot{m} C^*}{A_t}$$

$$A_t = \frac{\pi (0.028)^2}{4} = 6.158 \times 10^{-4} \text{ m}^2$$

$$p_c = \frac{(2.632)(1600)}{6.158 \times 10^{-4}}$$

$$\boxed{p_c = 6.835 \times 10^6 \text{ Pa}}$$

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Burning rate

$$r = 0.004p^{0.45}$$

$$r = 0.004(6.835 \times 10^6)^{0.45}$$

$$\boxed{r = 8.90 \text{ mm/s}}$$

$$r = 0.00890 \text{ m/s}$$

$$\dot{m} = \rho_p A_b r$$

$$A_b = \frac{\dot{m}}{\rho_p r}$$

$$\rho_p = 1540 \text{ kg/m}^3$$

$$A_b = \frac{2.632}{1540(0.00890)}$$

$$\boxed{A_b = 0.1921 \text{ m}^2}$$

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Propellant grain diameter

$$A_b = \frac{\pi D_p^2}{4}$$

$$D_p = \sqrt{\frac{4A_b}{\pi}}$$

$$D_p = \sqrt{\frac{4(0.1921)}{\pi}}$$

$$D_p = 0.4946 \text{ m}$$

$$\boxed{D_p \approx 495 \text{ mm}}$$

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MODYNAMICS

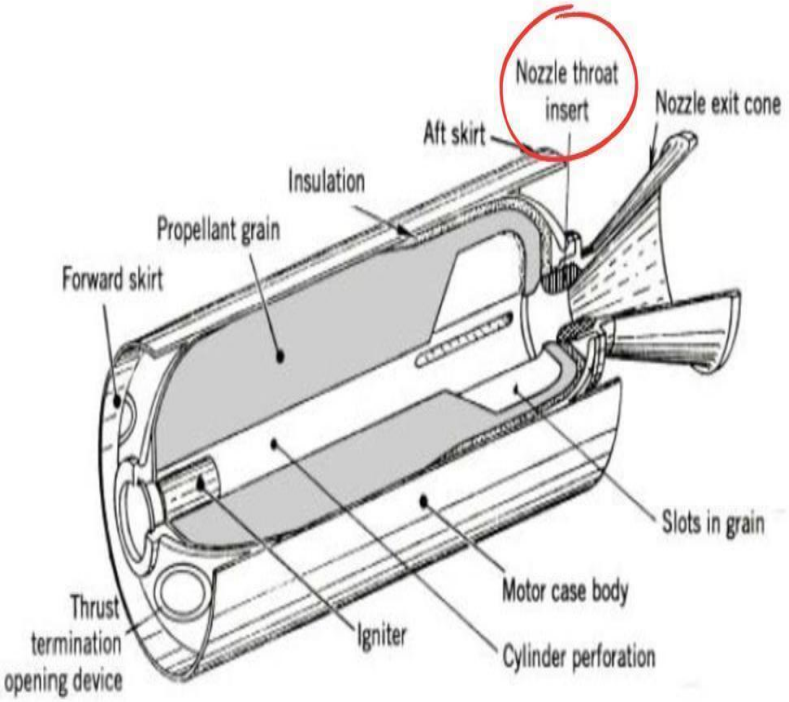
IGNITION PROCESS

- Solid propellant ignition comprises a series of complicated rapid events that start upon the receipt of a signal (usually electric) that ignites the propellant and this hot gas flows from the igniter to the motor grain surface,
- spreading of a flame over the entire burning surface area,
- filling the chamber cavity with gases, and elevating the chamber pressure without serious abnormalities such as
  - overpressures,
  - combustion oscillations,
  - damaging shock waves,
  - hangfires [delayed ignition],
  - flame extinguishment, and chuffing.

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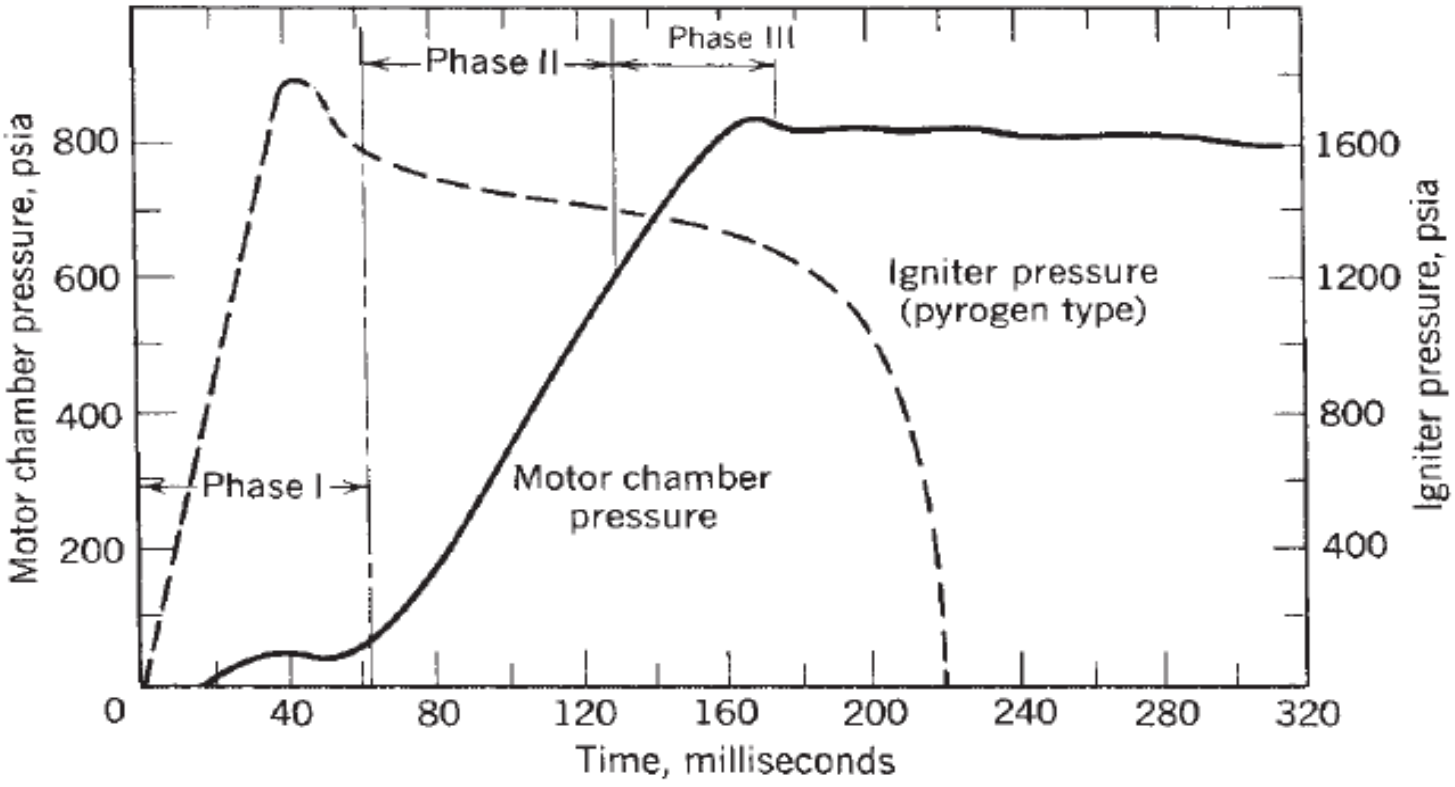
IGNITION PROCESS

- The *igniter* in a solid rocket motor is the component that heats and generates the gases necessary for motor ignition.
- Rocket motor ignition must usually be completed in fractions of a second for all but the very large motors.
- The pressure inside the motor rises to its equilibrium state in a very short time



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IGNITION PROCESS



IGNITION PROCESS

- Conventionally, the ignition process is divided into three phases for analytical purposes:

- **Phase I, *Ignition time lag*:** The period from the instant the igniter receives its signal until a portion of the grain surface burns and produces hot gases.

- **Phase II, *Flame-spreading interval*:** The time from first ignition of the grain surface until the complete grain burning area has been ignited.

- **Phase III, *Chamber-filling interval*:** The time for completing the chamber gasfilling process and for reaching equilibrium chamber pressure and mass flow.

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IGNITION PROCESS

- Ignition will be successful once sufficient grain surface is ignited and burning, so that the rocket motor will continue to raise its own pressure up to the operating chamber pressure.

- If the **igniter is not powerful enough**, some grain surfaces may burn only for a short time and the flame will be extinguished;

- the relevant critical processes seem to be located in a gas-phase reaction above the burning surface, where propellant vapors and/or decomposition products interact with each other and with the igniter gas products.

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IGNITION PROCESS

- Satisfactory attainment of equilibrium chamber pressures with full gas flows depends on

- (1) the characteristics of the igniter and its gas temperature, composition and issuing flow,

- (2) the motor propellant composition and grain surface ignitability,

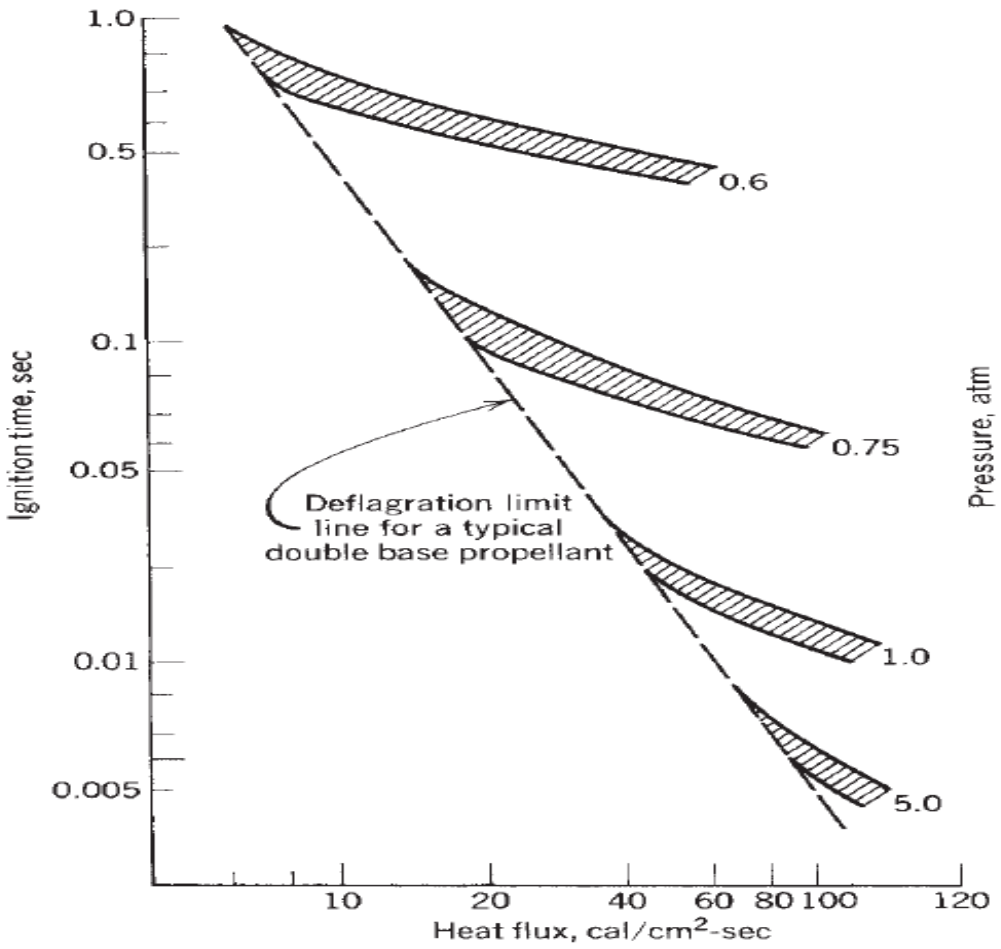
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IGNITION PROCESS

- (3) the radiation and convection heat transfer rates between the igniter gas and grain surface,
- (4) the grain flame spreading rate, and
- (5) the dynamics of filling the motor free volume with hot gas
- The **quantity and type of caloric energy** needed to ignite a particular motor grain in its prevailing environment has a direct bearing on most of the igniters' design parameters—particularly those affecting its required heat output.

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Propellant ignitability curves: effect of heat flux on ignition time



IGNITION PROCESS

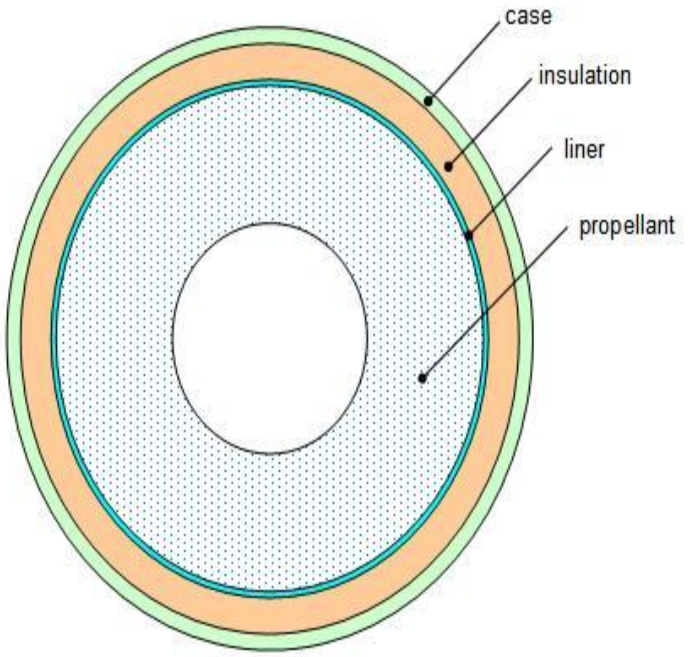
- Propellant ignitability is affected by several factors, including
- (1) propellant formulation,
- (2) propellant grain surface initial temperature,
- (3) surrounding pressure,
- (4) modes of heat transfer,
- (5) grain surface roughness,



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IGNITION PROCESS

- Propellant ignitability is affected by several factors, including
  - (6) propellant age,
  - (7) composition and hot-solid-particle content of igniter gases,
  - (8) igniter propellant and its initial temperature,
  - (9) velocity of hot igniter gases relative to the grain surface, and
  - (10) cavity volume and configuration.



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EXTINCTION OR THRUST TERMINATION

- Sometimes it is necessary to stop or extinguish burning in a solid propellant motor before all the propellant has been consumed.

- For example:
- 1. When a flight vehicle has reached the desired flight velocity (a ballistic missile attaining its predetermined velocity or a satellite achieving its desired orbit height), or when a precise total impulse cutoff is needed.

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EXTINCTION OR THRUST TERMINATION

- 2. As a safety measure, when it appears that a flight test vehicle will unexpectedly fly out of the safe boundaries of a flight test range facility.

- 3. To avoid collisions of stages during stage separation maneuvers (requiring a thrust reversal) for multistage flight vehicles.

- 4. During research and development, when the test requires examining a partially burned rocket motor.

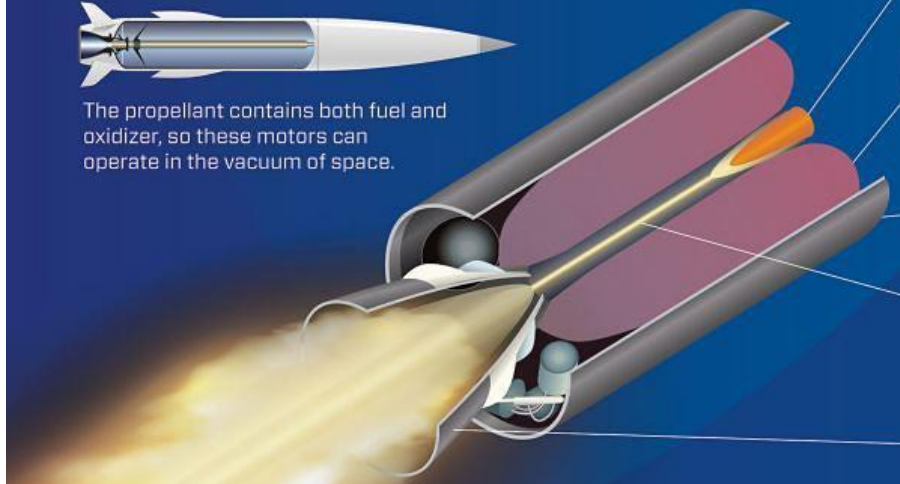
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NAMICS

How do Solid Rocket Motors work?

Aerojet Rocketdyne solid rocket motors incorporate advanced technologies and materials, including next generation propellants and lightweight motor cases, which improve performance and lower costs.

The propellant contains both fuel and oxidizer, so these motors can operate in the vacuum of space.



Igniter

An electrical signal is sent to the igniter, creating a flame that ignites the main propellant grain.



Propellant Grain

Baked to a pencil-eraser consistency

A mixture of fuel and oxidizer that is poured into a case and cured.

Motor Case

The body of the missile, it acts as a pressure vessel for the combustion chamber.

Propellant Burning Zone

As the propellant grain burns it produces high temperature combustion gases.

Nozzle

The combustion gases are accelerated through a nozzle, generating thrust to power a missile or rocket through the air or into space.

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EXTINCTION OR THRUST TERMINATION

- They are accomplished by highly predictable and reproducible explosive devices that suddenly open additional gas escape areas (thus causing pressure reduction) and neutralize the thrust by exhausting gases in a direction opposite to that of the motor nozzle.

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EXTINCTION OR THRUST TERMINATION

- Common mechanisms for achieving extinction are
- 1. **Very *rapid depressurization***, usually by a sudden and large increase of the nozzle throat area or by the fast opening of additional gas escape areas or ports.

- A most **common technique** neutralizes the net thrust or reverses its direction by suddenly opening exhaust ports in the forward end of the motor case.

- Such thrust reversals, using ports located on the forward bulkhead of the case, are achieved in the upper stages of Minuteman and Poseidon missiles.

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